MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 21: Tuesday 27 October 2026

(Week 11)

Remark. Q20.5 is leftover exercise from the Friday class in Week 10.

Remark. Q21.1 to Q21.5 (Spherical coordinates) are on Section 14.7 of Thomas' Calculus.

Q20.5. A steel beam with a pointed end and a round end is modelled as the set of points in the xyz-space in the cylinder of radius 1 centered at the origin, between the cone $z = x^2 + y^2 - 10$ and the hemisphere $z = \sqrt{1 - x^2 - y^2}$.

From the way that the steel beam is made, you deduce that its density $f(x, y, z)$ at a point (x, y, z) in the steel beam is proportional to the absolute value of its z-coordinate.

You also measured that the total mass of the steel beam is 5.

Find a formula for the density function f.

Q21.1. (a) Describe the surface S whose equation is given by $\rho = \sin\theta \sin\phi$ in spherical coordinates.

[Suggestion: First rewrite the equation in Cartesian coordinates.]

(b) Let D be the solid region bounded by S. Describe D in spherical coordinates.

Q21.2. Evaluate

$$\iiint_W e^{(x^2+y^2+z^2)^{3/2}}\, dx\, dy\, dz,$$

where W is the unit ball centered at the origin, i.e., $W = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\}$.

Q21.3. Evaluate the iterated integral

$$\int_{\pi/6}^{\pi/3} \int_{\csc\phi}^{2\csc\phi} \int_{0}^{2\pi} \rho^2 \sin\phi \, d\theta \, d\rho \, d\phi.$$

Q21.4. Evaluate the following integral:

$$\int_{-1}^{1} \int_{0}^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} \sqrt{x^2+y^2+z^2} \; dz \, dy \, dx.$$

Q21.5. Determine the volume of the region in the xyz-space bounded below by the cone $z^2 = x^2 + y^2$ and bounded above by the sphere of radius 1 centered at the origin.